Issuer Authorization Analytics & AI Insights — Write-Up

Executive Project Overview

1. Synthetic Data Design, Built-In Patterns & Assumptions

Developed a synthetic transaction dataset to simulate a realistic issuer authorization environment, covering different payment channels, merchant categories, acquirers, cardholder behaviours, and fraud scenarios.

Transaction Volume & Coverage

- Created a dataset of 220,000 transactions covering four months (July 1 – October 31, 2026), with a total transaction value of $41.45M.
- Simulated activity across 17,542 unique tokenized cardholders, 40 merchants, 10 Merchant Category Codes (MCCs), and 5 acquirers.

Merchant & Acquirer Performance

- Included different merchant risk profiles, from low-risk everyday transactions such as groceries and utilities to higher-risk categories such as cryptocurrency, luxury goods, and electronics.
- Simulated 5 different acquirer performance levels, with modern payment gateways achieving approximately 91–93% approval rates, compared with 80.8% for a legacy processor affected by switch timeouts and formatting issues.

Payment Channel & Security Trends

- POS Contactless and Chip & PIN transactions demonstrate higher approval rates (95.3% and 94.8%, respectively) with very low fraud levels.
- E-Commerce transactions show relatively lower approval (87.7%) and higher fraud incidence (3.65%).
- The dataset captures the impact of 3DS authentication and network/wallet tokenization, which improve approval rates and reduce fraud percentages

Seasonality & Fraud Behaviour

The dataset incorporates realistic transaction patterns to understand how customer behaviour and fraud activity change over time:

- Payday Patterns (25th–5th): Transaction volumes increase while insufficient-funds declines in post pay day compared with the pre-payday period.
- Late-Night Fraud Activity (01:00–05:00): Fraud incidence increases by approximately 3.6 times, simulating automated bot attacks and unusual overnight transaction activity.
- Cyber Retail Surge (October 15–25): Simulated a 35% increase in e-commerce transactions during promotional shopping periods, creating additional authorization and false-decline risks.
- Velocity-Based Fraud Attacks: Included compromised-card attack scenarios involving multiple rapid transaction attempts within short periods.

2. Machine Learning Approach, Validation & Threshold Business Impact

The primary objective was to build a machine learning model capable of predicting the likelihood of a transaction being fraudulent at authorization time. The goal was to improve fraud detection while reducing unnecessary declines of legitimate customer transactions.

Model Development & Validation

- Developed an XGBoost-based fraud prediction model using transaction attributes, cardholder spending behaviour, transaction velocity, and historical merchant-category and acquirer patterns.
- Followed a strict chronological validation approach to ensure the model is evaluated on future transactions rather than randomly distributed historical data.

The dataset was divided into three stages:

- **Training:** July 1 – September 15 (131,836 transactions).
- **Validation:** September 16 – October 5 (34,475 transactions), used for model tuning and threshold selection.
- **Out-of-Time Testing:** October 6 – October 31 (53,689 transactions), used to evaluate performance on unseen transactions, including the October promotional shopping surge.

Model Performance – Baseline vs. Advanced Model

Compared a Logistic Regression baseline against the advanced XGBoost model using the same out-of-time test dataset.

- **ROC-AUC:** Improved from 0.9145 to 0.9373.
- **PR-AUC:** Improved from 0.5138 to 0.6161, indicating better identification of fraudulent transactions.

The advanced model also incorporates **SHAP**-based explainability to identify the key factors influencing fraud predictions, including unsecured e-commerce transactions, unusual spending patterns, historical MCC fraud rates, and authentication signals.

Business Impact & Threshold Optimization

On the test dataset, the model delivered:

- Approximately **$573.9K in estimated net savings**.
- **$117.5K in incremental value** compared with legacy static fraud rules.
- Approximately **44.2% reduction in false declines**, from 1,102 to 615 transactions.

3. AI Insights Engine Architecture, Security & Grounding

Built an AI-powered insights engine to make the authorization dashboard more interactive and useful for business and operational teams.

Key Capabilities

- **Interactive AI Experience:** Users can ask business-related questions, explore follow-up insights, and use suggested questions to investigate specific trends.

- **Context-Aware Insights:** The AI engine receives structured, pre-calculated dashboard metrics covering approval rates, acquirer and merchant performance, decline patterns, 3DS failures, seasonality, and fraud model results.
- **Grounded AI Responses:** Python handles the underlying calculations before passing the results to the LLM. This ensures that responses are based on validated dashboard data rather than calculations performed independently by the language model.
- **Hallucination Controls:** Implemented strict prompting, low-temperature generation (0.1), and data-grounding checks to prevent unsupported responses.
- **Secure & Resilient Architecture:** API credentials are managed server-side, with a deterministic analytics fallback to keep the dashboard functional when external AI services are unavailable.

4. Production Roadmap

The current solution demonstrates the feasibility of combining transaction analytics, machine learning, and AI-driven insights. The next phase would focus on making the solution suitable for a live payment authorization environment.

Key Areas of Enhancement

- Real-Time Transaction Monitoring: Introduce low-latency streaming features using Redis/Feast to monitor transaction velocity and cardholder spending behaviour in real time
- Intelligent Risk-Based Authorization: Move beyond a simple approve-or-decline approach by introducing risk-based decision routing:
    - Low Risk: Approve the transaction.
    - Medium Risk ($0.08 \leq p < 0.12$): Trigger an additional 3DS authentication challenge.
    - High Risk ($p \geq 0.12$): Apply appropriate fraud prevention controls.
- Ongoing Model Monitoring: Establish continuous monitoring for model performance, data drift, fraud-pattern changes, and approval-rate variations to maintain reliability as transaction behaviour evolves.